

2025-10-31 - Valentini (10)

Previous lecture: [2025-10-30 - Valentini \(9\)](#).

Next lecture: [2025-11-06 - Valentini \(11\)](#).

Recap

$$\omega_R = \omega_{pe} \sqrt{1 + 3k^2 \lambda_{De}^2}$$

Langmuir waves

$$\left| \frac{\omega_I}{\omega_R} \right| \ll 1 \quad \text{and} \quad v_\phi = \frac{\omega_R}{k} \gg v_{th\alpha}$$

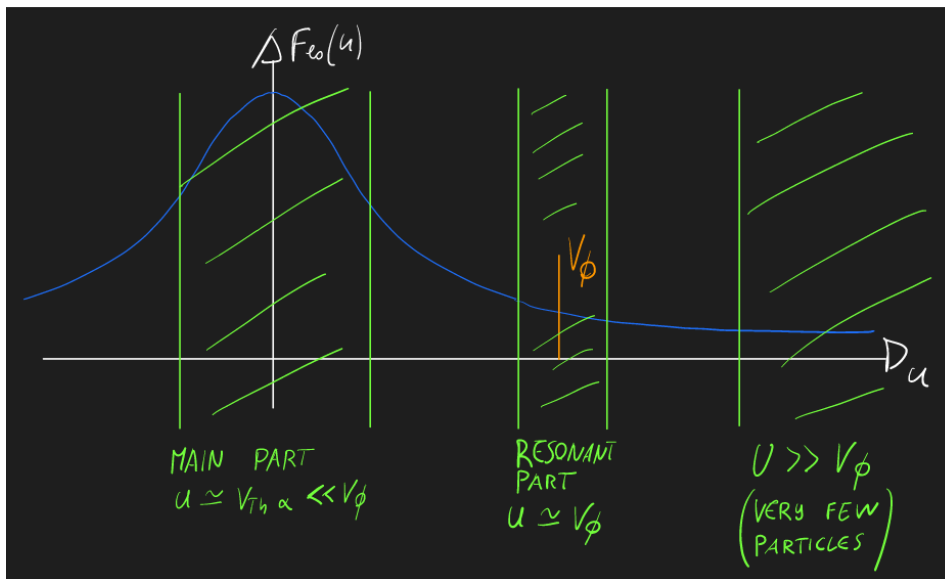
With the condition (yet to be discussed)

$$k^2 \lambda_{De}^2 \ll 1$$

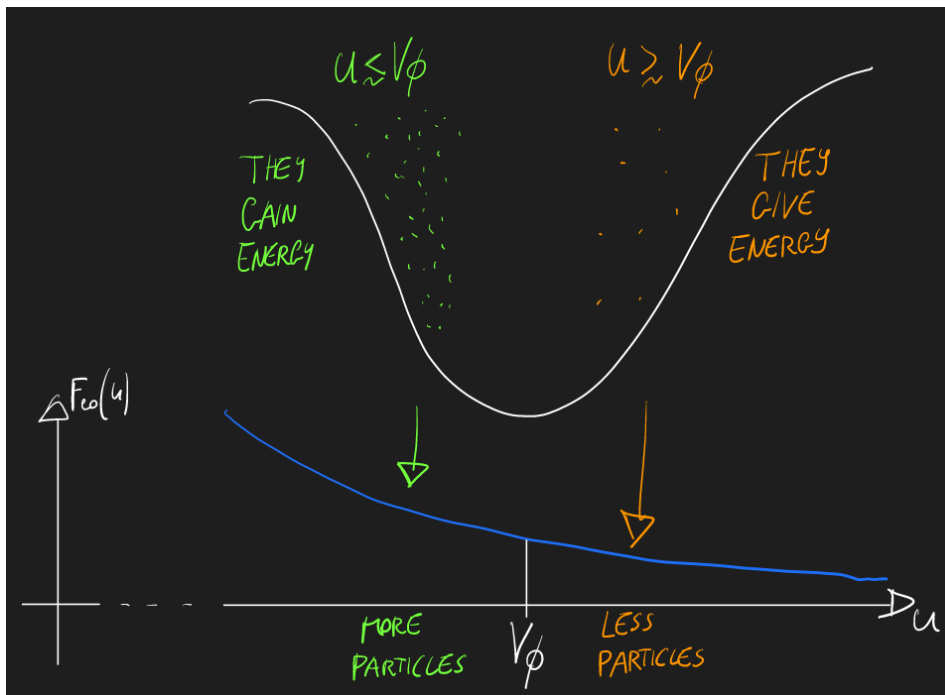
We got

$$\omega_I = \frac{\pi}{2} \frac{\omega_R^2}{k^2 n_{\alpha 0}} \left. \frac{\partial F_{\alpha 0}(u)}{\partial u} \right|_{\omega_R/k}$$

PIC resonant wave



PIC WaveSurferModel



Equation of motion for one electron:

$$m_e \frac{d^2}{dt^2} x = -eE_0 \sin(kx) \implies \frac{d^2}{dt^2} x + \underbrace{\frac{eE_0 k}{m_e}}_{\omega^2} \sin(kx) = 0$$

The electron moves with the wave

There is resonant wave-particle interaction

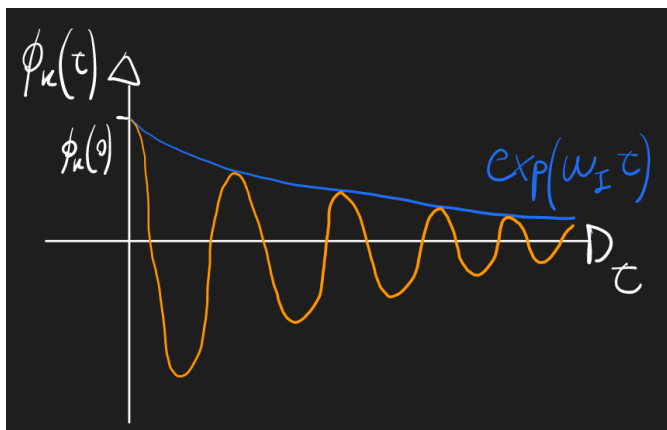
$$kx \ll 1 \implies \frac{d^2}{dt^2} x + \underbrace{\frac{eE_0 k}{m_e}}_{\omega^2} x = 0, \quad T = 2\pi \sqrt{\frac{m_e}{eE_0 k}}$$

with T being the trapping time, the period of oscillation of a charged particle in the potential hole. Particles are "trapped" and they oscillate between one wavefront and the other.

The Landau scenario is

$$\omega_I^{-1} < \tau$$

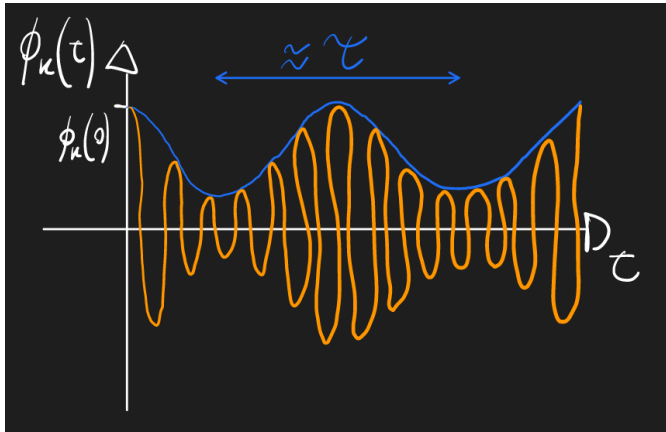
PIC landau scenario



The O'Neil scenario (which is non-linear) is

$$\omega_I^{-1} \approx \tau$$

PIC O'Neil scenario



$$\tau \approx \sqrt{\frac{m_e}{eE_0k}}$$

$$E(x) = E_0 \sin(kx)$$

$$E = -\frac{d}{dx}\phi \implies \phi = -\int E dx = \underbrace{\frac{E_0}{k}}_{\phi_0} \cos(kx)$$

The energy is

$$\mathcal{E} = \frac{1}{2}m_e(v - v_\phi)^2 - e\phi_0 \cos(kx) = \text{const}$$

we evaluate it at $v = v_\phi$ (where we also have $\phi = \phi_{\max}$)

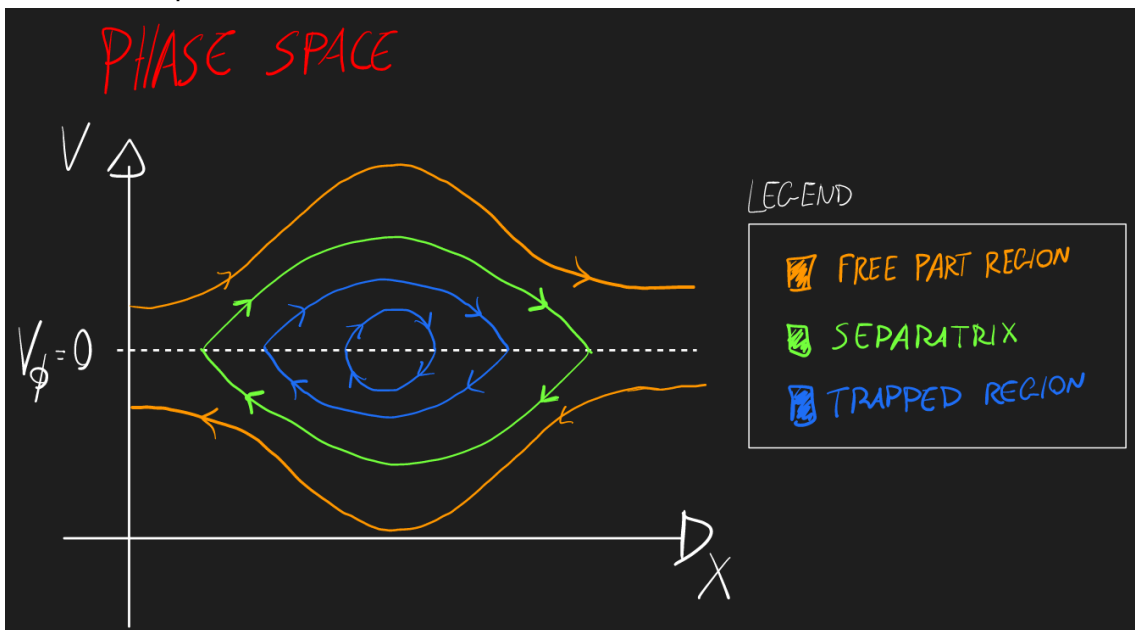
$$\mathcal{E} = -e\phi_{\max}$$

and

$$(v - v_\phi)^2 = \frac{2}{m_e}[-e\phi_{\max} + e\phi_0 \cos(kx)]$$

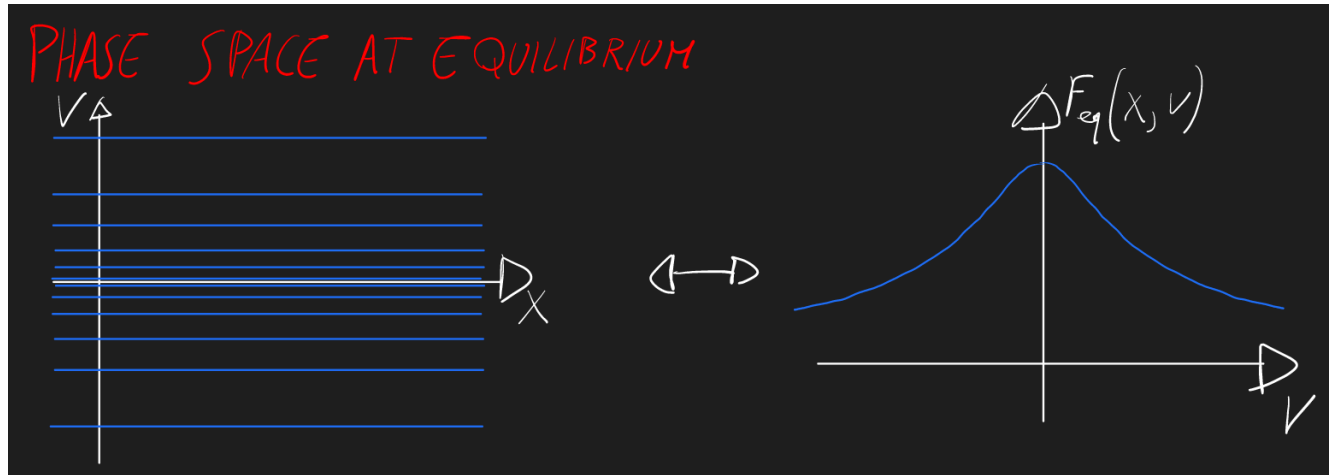
$$\implies v = v_\phi \pm \sqrt{\frac{2}{m_e}[-e\phi_{\max} + e\phi_0 \cos(kx)]}$$

PIC Phase space



If we plot the level lines of an unperturbed maxwel distribution f at $t = 0$, we see straight lines. These are closer to one another close to 0, detaching while moving away from the zero.

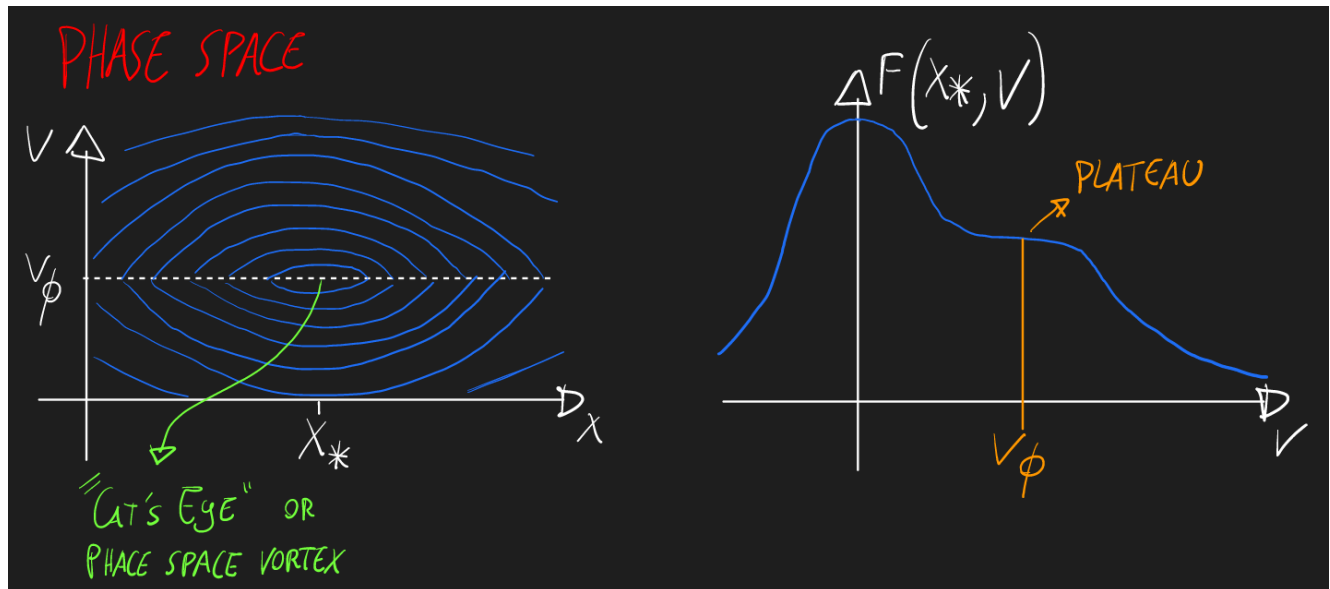
PIC maxwell lines - maxwell distribution



If we do a lateral cut, we obtain the usual form of the maxwell distribution (second picture)

We evolve the system at $t = t_{\max}$ in order to observe the nonlinear effect.

PIC nonlin phase space - plateau



$$F_{e0}(u) = \frac{n_0 m_e^{1/2}}{(2\pi k_B T_{e0})^{1/2}} \exp\left(-\frac{m_\alpha}{2k_B T_{e0}} u^2\right)$$

$$\frac{\partial F_{e0}}{\partial u} = \frac{n_0 m_e^{1/2}}{(2\pi k_B T_{e0})^{1/2}} \left(-\frac{m_\alpha u}{k_B T_{e0}}\right) \exp\left(-\frac{m_\alpha}{2k_B T_{e0}} u^2\right)$$

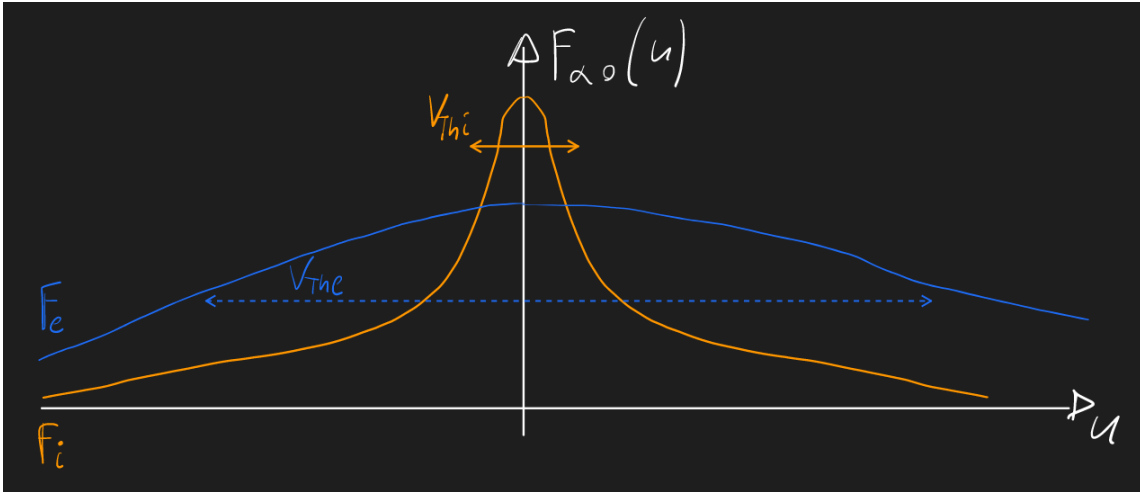
and (using $\omega_R \approx \omega_{pe}$ in the second passage)

$$\begin{aligned}
\omega_I &= -\frac{\pi}{2} \frac{n_\alpha}{k^2 n_0} \frac{m_e^{3/2}}{(2\pi)^{1/2} (k_B T_{e0})^{3/2}} \frac{\omega_R^4}{k} \exp\left(-\frac{m_e}{2k_B T_{e0}} \frac{\omega_R^2}{k^2}\right) = \\
&= -\sqrt{\frac{\pi}{8}} \frac{\omega_{pe} \omega_{pe}^3}{k^3 v_{th,e}^3} \exp\left[-\frac{m_e}{2k_B T_{e0}} \frac{1}{k^2} \omega_{pe}^2 (1 + 3k^2 \lambda_{De}^2)\right] = \\
&= -\sqrt{\frac{\pi}{8}} \frac{\omega_{pe}}{k^3 \lambda_{De}^3} \exp\left[-\frac{1}{2k^2 \lambda_{De}^2} (1 + 3k^2 \lambda_{De}^2)\right] = \\
&= -\sqrt{\frac{\pi}{8}} \frac{\omega_{pe}}{k^3 \lambda_{De}^3} \exp\left[-\frac{1}{2k^2 \lambda_{De}^2} - \frac{3}{2}\right]
\end{aligned}$$

Low frequency regime

$$\left| \frac{\omega_I}{\omega_R} \right| \ll 1, \quad v_{th,i} \ll \frac{\omega_R}{k} \ll v_{th,e}$$

PIC LowFreqRegime



$$D_R(k, \omega_R) = 0 \quad \implies \quad \omega_R = \omega_R(k)$$

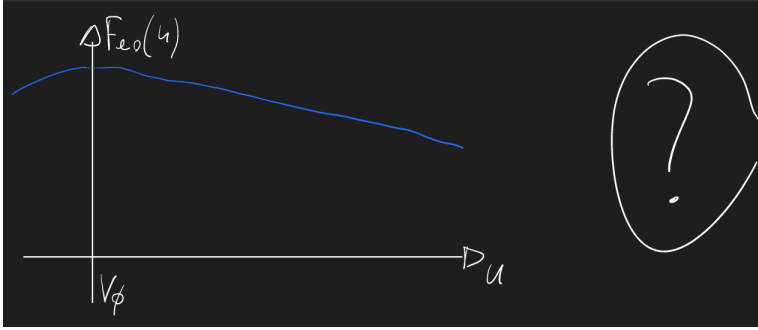
$$\begin{aligned}
0 = D_R(k, \omega_R) &= 1 - \sum_{\alpha} \frac{\omega_{p\alpha}^2}{k^2 n_{\alpha 1}} P \int_{-\infty}^{\infty} \frac{\partial F_{\alpha 0}(u)/\partial u}{u - \omega_R/k} du = \\
&= 1 - \frac{\omega_{pi}^2}{k^2 n_0} P \int_{-\infty}^{\infty} \frac{\partial F_{i0}(u)/\partial u}{u - \omega_R/k} du - \frac{\omega_{pe}^2}{k^2 n_0} P \int_{-\infty}^{\infty} \frac{\partial F_{e0}(u)/\partial u}{u - \omega_R/k} du
\end{aligned}$$

We already solved the first integral for ions, so we substitute the previous result.

We also say, for the second integral, that the majority of particles has $u \gg \omega_R/k$

$$0 = D_R(k, \omega_R) = 1 - \frac{\omega_{pi}^2}{k^2 n_0} \frac{n_0 k^2}{\omega_R^2} \left[1 + 3 \frac{k^2 \lambda_{Di}^2 \omega_{pi}^2}{\omega_R^2} \right] \underbrace{\left[-\frac{\omega_{pe}^2}{k^2 n_0} P \int_{-\infty}^{\infty} \frac{1}{u} \frac{\partial F_{e0}(u)}{\partial u} du \right]}_{\mathcal{I}}$$

PIC electrons



We can solve the second integral as follows:

first we substitute $F_{e0}(u)$ and derive

$$\mathcal{I} = -\frac{\omega_{pe}^2}{k^2} P \int_{-\infty}^{\infty} \frac{1}{u} \left[-\frac{n_0 m_e^{3/2}}{(2\pi)^{1/2} (k_B T_{e0})^{3/2}} \cdot \exp\left(-\frac{m_e}{2k_B T_{e0}} u^2\right) \right] du$$

then we use the substitution

$$y^2 = u^2 \frac{m_e}{2k_B T_{e0}} \quad \Rightarrow \quad y = \frac{m_e^{1/2}}{(2k_B T_{e0})^{1/2}} u$$

and

$$\begin{aligned} \mathcal{I} &= \frac{\omega_{pe}^2}{k^2} \frac{m_e^{3/2}}{(2\pi)^{1/2} (k_B T_{e0})^{3/2}} \underbrace{\int_{-\infty}^{\infty} \exp(-y^2) dy}_{\sqrt{\pi}} \sqrt{\frac{2k_B T_{e0}}{m_e}} = \frac{1}{k^2 \lambda_{De}^2} \\ \Rightarrow D_R(k, \omega_R) &= 1 - \frac{\omega_{pi}^2}{\omega_R^2} \left[1 + 3 \frac{k^2 \lambda_{Di}^2 \omega_{pi}^2}{\omega_R^2} \right] + \frac{1}{k^2 \lambda_{De}^2} = 0 \end{aligned}$$

We solve for ω_R

$$\Rightarrow \left(1 + \frac{1}{k^2 \lambda_{De}^2} \right) \omega_R^4 - \omega_{pi}^2 \omega_R^2 - 3k^2 \lambda_{Di}^2 \omega_{pi}^4 = 0$$

and

$$\begin{aligned} \omega_R^2 &= \frac{\omega_{pi}^2 \pm \sqrt{\omega_{pi}^4 + 12k^2 \lambda_{Di}^2 \omega_{pi}^4 \left(\frac{1+k^2 \lambda_{De}^2}{k^2 \lambda_{De}^2} \right)}}{2 \cdot \left(\frac{1+k^2 \lambda_{De}^2}{k^2 \lambda_{De}^2} \right)} = \\ &= \frac{1}{2} \frac{k^2 \lambda_{De}^2}{1 + k^2 \lambda_{De}^2} \left[\omega_{pi}^2 \pm \omega_{pi}^2 \sqrt{1 + 12 \frac{T_{i0}}{T_{e0}} (1 + k^2 \lambda_{De}^2)} \right] = \\ &= \frac{\omega_{pi}^2}{2} \frac{k^2 \lambda_{De}^2}{1 + k^2 \lambda_{De}^2} \left[1 \pm \sqrt{1 + 12 \frac{T_{i0}}{T_{e0}} (1 + k^2 \lambda_{De}^2)} \right] \approx \\ &\approx \frac{\omega_{pi}^2}{2} \frac{k^2 \lambda_{De}^2}{1 + k^2 \lambda_{De}^2} \left[1 \pm \left(1 + 6 \frac{T_{i0}}{T_{e0}} + \dots \right) \right] \end{aligned}$$

Where we used

$$\lambda_{Di}^2 = \frac{k_B T_{i0}}{4\pi n_0 e^2} \frac{T_{e0}}{T_{e0}} = \lambda_{De}^2 \frac{T_{i0}}{T_{e0}}$$

Now we only keep the (+) solution in order to avoid imaginary ω_R , and we obtain

$$\omega_R = \pm \frac{\omega_{pi} k \lambda_{De}}{(1 + k^2 \lambda_{De}^2)^{1/2}} \left(1 + \frac{3T_{i0}}{T_{e0}} \right)^{1/2}$$

and we also find the ion sound speed as

$$\omega_{pi} \lambda_{De} = \left[\frac{\cancel{4\pi n_0 e}}{m_i} \frac{k_B T_{e0}}{\cancel{4\pi n_0 e}} \right]^{1/2} = c_s$$

so

$$\omega_R = \pm \frac{k c_s}{\sqrt{1 + k^2 \lambda_{De}^2}} \sqrt{1 + 3 \frac{T_{i0}}{T_{e0}}}$$

which are the ion acoustic waves.

and we can affirm that, for $k^2 \lambda_{De}^2 \ll 1$

$$\left| \frac{\omega_I}{\omega_R} \right| \ll 1 \quad \Rightarrow \quad \frac{T_{i0}}{T_{e0}} \ll 1$$

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